

CE 310 — Week 6 In-Class Exercise: Empirical Percentiles Across CE Practice

Total: 100 points (Section A: 16 · Section B: 60 · Chart 1: 7 · Chart 2: 8 · Memo: 9)

Before You Begin

Dataset 1: Week5_CE_Measurements.csv — the same file from Week 5.

Columns (Dataset 1): A=RecordID · B=Supplier (“Alpha”/“Beta”) · C=TimeOfDay (“AM_Peak”/“PM_Peak”) · D=fc_psi · E=Speed_mph · F=CostPerCY

Dataset 2 (new): Week6_Noise_Measurements.csv — A=RecordID · B=Location (“Roadside”/“Courtyard”) · C=NoiseLevel_dB

Answer sheet: the Week 6 in-class answer sheet — sheets Answers, Work, Charts and Written. Do not rename, reorder, or delete any sheet; adding your own is fine and expected. The assignment has its own file, the Week 6 assignment answer sheet.

Cell layout on the Answers sheet: Column A = question # (locked) · Column B = question text (locked) · **Column C = your formula/approach** · **Column D = intermediate value** (★ questions only) · **Column E = your numerical answer** · Column F = unit (locked).

Type your formula as text. In Col C, put an apostrophe before the = so Excel stores the formula instead of running it — for example, '=AVERAGE(B2:B8761). Without the apostrophe Excel evaluates it and your method is lost. **Screenshots are not accepted;** the typed formula is what earns the approach credit.

Background — From Week 5 to Week 6

Last week you fit a Normal model (concrete, speed) or lognormal model (cost) to this same dataset and used NORM.INV/LOGNORM.DIST to estimate percentiles. This week you compute the *actual* empirical percentiles — no model, no assumption about shape — and compare.

Setup (≈10 min): You do all of this week’s work in **one file: your Answer Sheet**. Add two sheets to it and import one dataset onto each, starting at A1:

1. **Data** — Week5_CE_Measurements.csv. Same file *and* same sheet name as Week 5, so you can copy the sheet straight across from your Week 5 file if you still have it. Column E is Speed_mph; the percentile table in Section B reads it as Data!\$E\$2:\$E\$501, so the sheet has to carry that exact name.
2. **Noise** — Week6_Noise_Measurements.csv, the new one.

Confirm =COUNTA(A:A)–1 returns **500** on each, and confirm your column letters match the tables above. Only Instructions, Answers, Work, Charts and Written have to keep their names — adding Data and Noise is expected. Keeping the data in the same file is what lets you build the charts where they are graded.

Excel Quick Reference

Concept	Formula
kth largest / smallest	=LARGE(range, k) / =SMALL(range, k)
Median	=MEDIAN(range)
Interpolated percentile	=PERCENTILE(range, p)
Exceedance probability	=COUNTIF(range, ">x") / 500 * 100
Band probability (between two values)	=COUNTIFS(range, ">lower", range, "<upper") / 500 * 100
Conditional percentile	=PERCENTILE(FILTER(range, condition), p)
Total exposure above threshold	=SUMPRODUCT((range-threshold)*(range>threshold))

Section A — Concrete Strength and Site Acoustics (16 pts)

Section A drops the model. Every answer comes from ranking the data and reading off a position, which is what lets you see what Week 5's fitted distribution added — and what it assumed.

Same four-question toolkit — LARGE, SMALL, MEDIAN, COUNTIF — applied to two independent domains. Round to the nearest 0.01 unless the question says otherwise.

Concrete (Dataset 1, column D `fc_psi`, all 500 records):

A1 (2 pts) — 90th Percentile Concrete Strength

Compute using the exact-rank method. *Hint: LARGE, $k = \text{ROUND}(500 \times 0.10, 0)$.*

A2 (2 pts) — 10th Percentile Concrete Strength

Compute using the exact-rank method — this is the value ACI 318 cares about. *Hint: SMALL, same k as A1.*

A3 (2 pts) — Percentage Below 4,000 psi

What percentage of all 500 tests fall below 4,000 psi? *Hint: COUNTIF, remember to divide by 500 and multiply by 100.*

A4 (2 pts) — Empirical Median Concrete Strength

The median of all 500 records.

Site acoustics (Dataset 2, column C `NoiseLevel_dB`, all 500 records):

A5 (2 pts) — L10 Noise Level (Peak)

The level exceeded 10% of the time, using the exact-rank method. *Hint: LARGE, same k logic as A1 — L10 is the 90th percentile of the readings.*

A6 (2 pts) — L90 Noise Level (Background)

The level exceeded 90% of the time, using the exact-rank method. *Hint: SMALL, same k logic as A2 — L90 is the 10th percentile of the readings.*

A7 (2 pts) — L50 Noise Level (Empirical Median)

The median of all 500 readings.

A8 (2 pts) — Percentage Above the 65 dB HUD Threshold

What percentage of all 500 readings exceed 65 dB — the HUD site-acceptability threshold? *Hint: COUNTIF, same pattern as A3.*

Section B — Traffic Speed and Construction Cost (60 pts)

Section B runs the two approaches side by side on speed and cost, and asks which number you would defend to a client.

Traffic Speed (column E) — all 500 records unless a subgroup is specified

B1 (4 pts) — 85th Percentile Speed, All Records

Use the exact-rank method — this is the value the 85th-percentile speed rule (Part 1 of this week's lecture) is built on. *Hint: LARGE, $k = \text{ROUND}(500 \times 0.15, 0)$.*

B2 ★ (5 pts) — 10th Percentile Speed, All Records

Use the exact-rank method. **Column D:** what k did you use? *Hint: SMALL.*

B3 (4 pts) — Gap From Week 5's Normal Model

Week 5's Normal model estimated the 85th-percentile speed at 62.5 mph. What is the gap between your B1 answer and 62.5 ($B1 - 62.5$)?

B4 (4 pts) — Percentage Above 65 mph

What percentage of all records show speed above 65 mph?

B5 (4 pts) — Q1 (25th Percentile) Speed

The first quartile across all records.

B6 (4 pts) — Q3 (75th Percentile) Speed

The third quartile across all records.

IQR diagnostic (ungraded): Compute $IQR = B6 - B5$. By the Normal approximation, $IQR \div 1.35 \approx \sigma$. Compare this robust estimate to your Week 5 standard deviation (~7.8 mph for all records). Close agreement means the distribution is approximately Normal through the middle; a larger gap suggests the tails depart from the Normal model.

B7 ★ (5 pts) — 85th Percentile Speed, AM Peak Only

Restrict to **AM Peak** records. **Column D**: how many AM Peak records are there? (Use COUNTIF on column C.) *Hint: LARGE(FILTER(. . .)) or the array-formula equivalent.*

B8 (4 pts) — 85th Percentile Speed, PM Peak Only (Interpolated)

Restrict to **PM Peak** records and use the interpolated method. *Hint: PERCENTILE(FILTER(. . .)).*

B9 (4 pts) — Percentage Above 45 mph

What percentage of all records show speed above 45 mph? (Week 5's Normal model estimated 88.5% for this same question — you are not required to enter the comparison, just the empirical value.)

Construction Cost (column F) — the AACE contingency question

B10 ★ (5 pts) — 90th Percentile Cost (Exact Rank)

Use the exact-rank method — this is the value AACE-style contingency budgeting is built on. **Column D**: what k did you use? *Hint: LARGE, k = ROUND(500 × 0.10, 0).*

B11 (4 pts) — 90th Percentile Cost (Interpolated)

Use the interpolated method, then compare to B10. *Hint: PERCENTILE.*

B12 (4 pts) — Gap From Week 5's Lognormal Model

Week 5's lognormal model estimated the 90th-percentile cost at \$289.61/CY. What is the gap between your B10 answer and 289.61 (B10 – 289.61)?

B13 ★ (5 pts) — Total Dollar Exposure Above \$250/CY

The total across all records whose cost exceeds \$250/CY — i.e., $\sum \max(\text{Cost} - 250, 0)$. **Column D**: how many records exceed \$250? *Hint: SUMPRODUCT.*

B14 (4 pts) — Median Cost, Supplier Beta Only

Restrict to **Supplier Beta** records. *Hint: MEDIAN(FILTER(. . .)).*

Where the working goes. Your Answer Sheet has a **Work** sheet with a labelled, pre-filled block for each table below. Build them there and nowhere else — the charts are scored by looking at the cells they plot, so a table in a blank corner of another sheet is a table your preceptor has to hunt for.

Required Charts (15 pts)

Both charts belong on the **Charts sheet** of your Answer Sheet, in the outlined boxes. That sheet lists what each chart is graded on.

Build the chart wherever your data is, then right-click it -> **Move Chart** -> **Object in** -> Charts, and drag it into the matching box. Leave it as a live chart — do not replace it with a picture of itself. Your preceptor clicks the chart to see which cells it plots, which is how a correct chart built from the wrong column gets caught.

Chart 1 (7 pts) — Concrete Strength Duration Curve

This one goes on your **Data sheet**, beside the concrete data — it is 500 rows long and lines up with the dataset row for row. In the first two empty columns (H and I if your import started in A1) build Rank 1–500 and a sorted strength column, =LARGE(\$D\$2:\$D\$501, H2) filled down. Plot Rank on the x-axis and strength on the y-axis as a line chart. Title: “Concrete Compressive Strength Duration Curve (n = 500)”. Add a horizontal reference line at your A2 answer (the ACI 318 threshold).

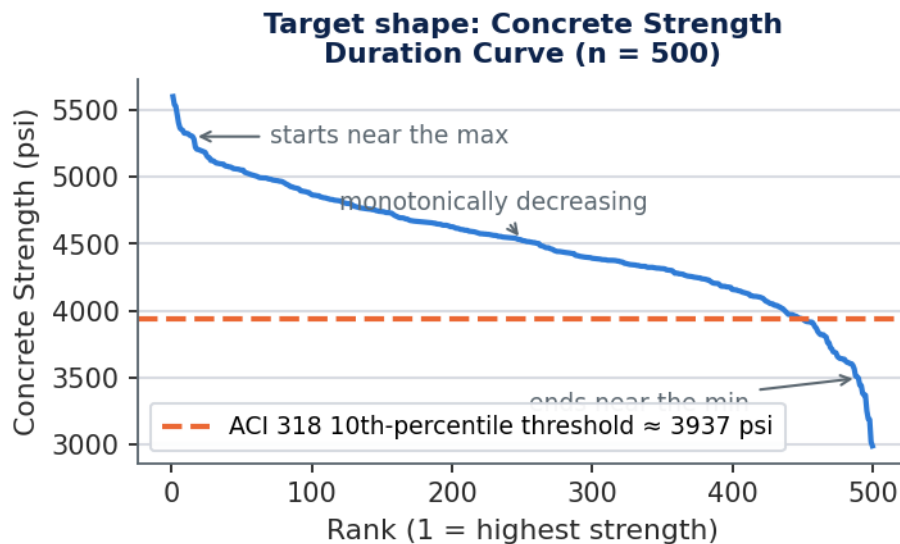


Figure 1: Target shape for Chart 1

Chart 2 (8 pts) — Empirical vs. Normal Model, Speed

This one goes on the **Work sheet**, where the 99 percentiles from 0.01 to 0.99 are already filled down column A, rows 7 to 105. Fill the two yellow columns:

B7: =PERCENTILE(Data!\$E\$2:\$E\$501, \$A7) (empirical)
 C7: =NORM.INV(\$A7, 54.41, 7.81) (Normal model)

Fill both down to row 105, then plot B7:C105 against A7:A105 as a scatter with smooth lines, both series, with a legend. Title: “Vehicle Speed: Empirical vs. Normal Model”.

Use your own sheet name and column letters in that first formula if your speed data did not land in column E.

Memo Deliverable (9 pts, in the Written sheet)

Write 3 short paragraphs:

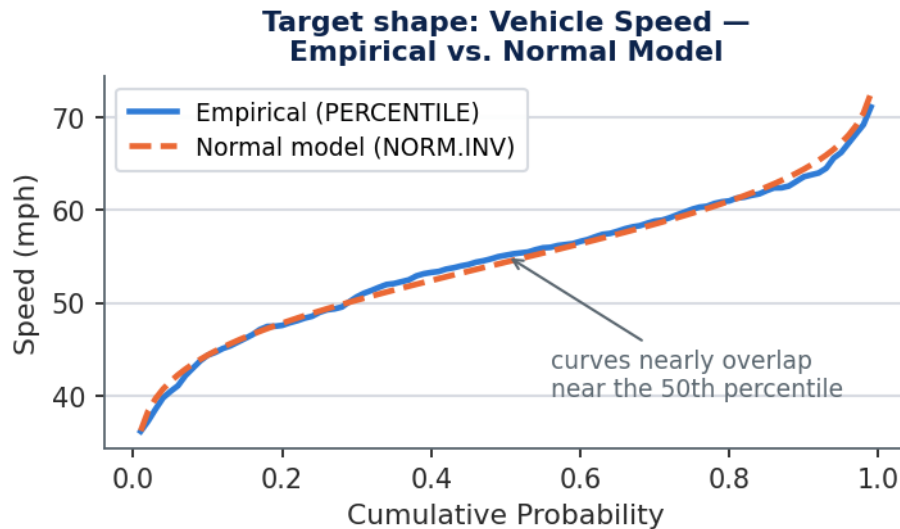


Figure 2: Target shape for Chart 2

1. **Paragraph 1 (3 pts).** Report your A2 (empirical 10th-percentile concrete strength) and compare it to Week 5's Normal-model estimates for Supplier Alpha (4,191 psi) and Supplier Beta (3,747 psi) individually — cite the actual Alpha/Beta empirical values from lecture (4,181 / 3,688 psi) and state which supplier(s) would fail a 4,000 psi spec.
2. **Paragraph 2 (3 pts).** Report your B13 (total cost exposure above \$250/CY) and explain what it tells a project manager that the B10/B11 percentile value alone does not.
3. **Paragraph 3 (3 pts).** Compare your B1 (all-records 85th-percentile speed) to B7 and B8 (AM/PM conditional). Cite at least 2 numbers. What would you recommend a traffic engineer do differently for AM vs. PM if this were a real corridor?

Submission: Save as CE310_W6_<NetID>.xlsx and upload to the Week 6 D2L dropbox. Enter your NetID in the **NetID** cell at the top of the Answers sheet — that cell is what matches your work to your student record. A 5-point deduction applies if it is blank.

Before You Submit

- ☐ Full name and NetID entered on the Answer Sheet — a blank NetID cell takes a 5-point deduction
- ☐ File saved as .xlsx and named CE310_W6_<NetID>.xlsx
- ☐ Sheet tabs not renamed, reordered, or deleted
- ☐ Every answer typed into its designated cell in **Col E** — answers placed anywhere else receive no credit
- ☐ Your formula typed as text in **Col C**, with an apostrophe before the = — screenshots are not accepted
- ☐ ★ questions also have their intermediate value in **Col D**
- ☐ Both charts sitting in their boxes on the Charts sheet, as live charts and not pictures, with titles and axis labels
- ☐ Memo written in the Written sheet
- ☐ Uploaded to D2L by **9:00 AM next Wednesday**

Grading Summary: Section A (16) + Section B (60) = 76. Chart 1 (7) + Chart 2 (8) + Memo (9) = 24.
Total: 100 pts.